A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.8: Ethernet and CSMA/CD

Total Questions: 6

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w25_qp_12 (Q5.d)

5 A local area network (LAN) has five computers, one switch and one server.

(d) Another type of network is a bus network.

Ethernet is used to transmit and receive data between the devices on the bus network.

Describe how collisions are detected and managed on this network.

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.... [3]

Question 2 | Source: 9618_s24_qp_13 (Q5.c.i)

5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(c) A different part of the network uses the Ethernet protocol.

(i) A collision is detected.

Describe how the collision is managed using Carrier Sense Multiple Access/Collision Detection (CSMA/CD).

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.... [2]

Question 3 | Source: 9618_s24_qp_13 (Q5.c.ii)

5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(c) A different part of the network uses the Ethernet protocol.

(ii) Identify **two** drawbacks of using CSMA/CD.

1 ....

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2 ....

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[2]

Question 4 | Source: 9618_w23_qp_12 (Q7.c)

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(c) State **three** tasks performed by devices to deal with collisions when using the Carrier Sense Multiple Access / Collision Detection (CSMA/CD) protocol in a network.

1 ....

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2 ....

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3 ....

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[3]

Question 5 | Source: 9618_s23_qp_12 (Q1.d)

1 A company has a LAN (local area network).

(d) Computers can be connected using Ethernet.

Describe what is meant by **Ethernet**.

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.... [3]

Question 6 | Source: 9618_w22_qp_11 (Q8)

8 A Local Area Network (LAN) uses a bus topology.

Describe how Carrier Sense Multiple Access / Collision Detection (CSMA / CD) is used in a bus network.

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.... [4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q5.d)

5(d)	**1 mark** per bullet point, **max 3** marks • The use of <u>CSMA/CD</u> // Carrier Sense Multiple Access with Collision Detection • The workstations listen to the communication channel • ... and send data only when there is no data being transmitted / the line is idle • If a collision is detected, transmission is aborted • …and a jamming signal is sent • The workstation calculates a random wait time before trying to re-transmit • The random time is increased if there are multiple collisions	**3**

Mark Scheme for Question 2 | Source: 9618_s24_ms_13 (Q5.c.i)

5(c)(i)	**1 mark** each to **max 2**: • Jamming signal is transmitted by the **sending device** • Transmission is aborted • The **sending device** waits a **random** time before trying to send data again ... • ... if further collisions occur the wait time is increased	**2**

Mark Scheme for Question 3 | Source: 9618_s24_ms_13 (Q5.c.ii)

5(c)(ii)	**1 mark** each to **max 2**: • Random time increased each time so can be infinite waiting • May be constant jamming signal so nothing ever sends • Certain nodes cannot be prioritised • High power consumption • Only suitable for short distance network // limited distance • Not scalable // more nodes means exponentially longer waiting times	**2**

Mark Scheme for Question 4 | Source: 9618_w23_ms_12 (Q7.c)

7(c)	**1 mark** for each bullet point (**max 3**) • To monitor the communications channel • To send data only when there is no data being transmitted / the line is quiet / idle • **To detect a collision** and then stop transmissions of further data // transmit a **jamming** signal • To calculate a **random** wait time / back-off time • ... then retransmit the data after that random wait time • Increase **random** time if multiple collisions	**3**

Mark Scheme for Question 5 | Source: 9618_s23_ms_12 (Q1.d)

1(d)	**1 mark** each to **max 3** • A protocol (suite) • For **data transmission** over standard / universal **wired** / **cabled** network connections • Uses Carrier Sense Multiple Access / Collision Detection (CSMA/CD) • Data is transmitted in frames • … each frame has a source and destination (IP/MAC) address • ... and error checking data (so damaged frames can be resent)	**3**

Mark Scheme for Question 6 | Source: 9618_w22_ms_11 (Q8)

8	**1 mark** for bullet point (**max 4**): • CSMA/CD is a protocol used to detect and prevent collisions in a bus topology • before transmitting, a **device** checks if the channel is busy • If the channel is busy the device waits // if the channel is free the data is sent • because there is more than one computer connected to the same transmission medium • ... two workstations can start to transmit at the same time, causing a collision • If a collision is detected by the device, transmission is aborted / a jamming signal is transmitted • both devices wait a (different) **random** time and then try again	**4**